Transmission Bottlenecks and Battery Storage Mitigation for Spatially Clustered EV Charging

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1. Motivation and Research Question

Electric vehicles (EVs) can reduce direct transportation emissions, but the associated charging demand must be delivered through a power system with finite generation, storage, and transmission capacity. Well-to-wheel and life-cycle studies show that EV climate benefits depend on the upstream electricity supply rather than tailpipe emissions alone [1, 2, 3]. Average grid-emission factors are insufficient when charging load changes marginal generation, renewable curtailment, and power transfers across constrained networks [4, 5].

Chapter 2 of the dissertation proposal establishes a balancing-area consequential-emissions framework using a modified Grid Optimized Operation Dispatch (GOOD) model. However, aggregate regional modeling can hide an important operational mechanism: EV charging is spatially clustered, while clean generation is often remote. When transmission corridors bind, low-carbon imports can be curtailed and local fossil generation may be dispatched out of merit [6, 7]. This study asks:

How much of the consequential emissions from EV charging is attributable to transmission bottlenecks, and can strategically located battery energy storage systems (BESS) reduce this congestion-related emissions penalty?

2. Background and Contribution

Prior EV-grid studies increasingly combine empirical charging profiles with dispatch or capacity-expansion models to estimate marginal power-sector impacts [8, 9, 10, 11]. Storage has also been proposed as a mitigation resource for charging-related grid stress because it can shift renewable generation into high-demand charging periods and reduce dependence on constrained imports [12, 13, 14, 15]. Yet the emissions value of storage depends on where it is connected and whether it relieves a binding delivery constraint rather than merely shifting energy temporally.

The contribution of this extended abstract is a spatially resolved extension of the Chapter 2 GOOD framework. The proposed model links multi-day EV charging demand to transmission-constrained grid dispatch and evaluates three counterfactuals: no EV charging, EV charging without new BESS, and EV charging with endogenous BESS deployment. The key output is a congestion-emissions penalty: the difference between EV-attributable emissions under constrained transmission and a relaxed-transfer benchmark.

3. Data and Methods

EV Charging Demand

The demand module builds from the multi-day SOC-aware charging simulation. Vehicles are assigned heterogeneous battery capacities, daily vehicle miles traveled, home/work charging access, and charging-frequency preferences. Energy demand accumulates over multiple days, and charging is triggered when accumulated demand approaches usable battery capacity or when a preferred charging interval is reached. Each event is matched to an empirical charging-session pool by location type, charger level or housing type, and energy bin; the empirical session supplies start hour, end hour, power, and energy.

The current California aggregate case follows Chapter 2: approximately 2.5 million light-duty EVs, 35 miles/day, 0.3 kWh/mile, and charging-energy shares of 68% residential L1/L2, 4% workplace L2, 8% public L2, and 20% DC fast charging [16]. For the Chapter 3 extension, annual or representative-week charging energy is also accumulated by transportation analysis zone (TAZ). Home charging is assigned to the vehicle's home TAZ, while work and public charging are assigned using destination proxies until trip-level destinations are fully integrated. TAZ loads will be mapped to substations or transmission nodes using service territories, nearest-network assignment, or load-zone aggregation.

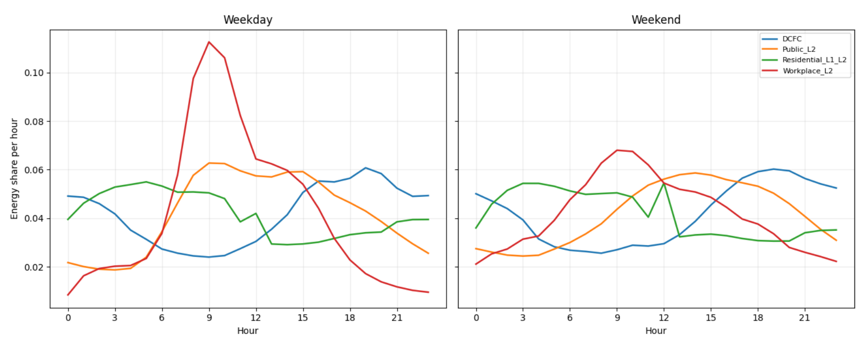

Figure 1: Empirical-session-based EV charging profiles by charger category and day type. These profiles provide the temporal load shape for the transmission-constrained GOOD extension.

Transmission-Constrained GOOD Extension

The Chapter 2 GOOD model minimizes operating and capacity-expansion cost subject to power balance, generator limits, ramping, storage inventory, policy constraints, and transmission exchange. Chapter 3 adds explicit transmission constraints at nodes $n \in N$ and lines $\ell = (i, j) \in L$. A simplified DC-flow formulation is:

$$f_{\ell,t} = b_\ell(\theta_{i,t} - \theta_{j,t}),$$

$$-\bar{F}_\ell - z_\ell^T \leq f_{\ell,t} \leq \bar{F}_\ell + z_\ell^T,$$

where $f_{\ell,t}$ is line flow, b_ℓ is susceptance, $\theta_{i,t}$ and $\theta_{j,t}$ are voltage angles, $\bar{F}_\ell$ is existing transfer capacity, and z_ℓ^T is optional line expansion.

Nodal power balance is:

$$\sum_{a \in A_n} g_{a,t} + \sum_{b \in B_n} d_{b,t}^{BESS} + \sum_{\ell \in In(n)} f_{\ell,t} = L_{n,t}^{base} + L_{n,t}^{EV} + \sum_{b \in B_n} c_{b,t}^{BESS} + \sum_{\ell \in Out(n)} f_{\ell,t} + w_{n,t},$$

where $g_{a,t}$ is generation, $L_{n,t}^{EV}$ is mapped EV charging load, $c_{b,t}^{BESS}$ and $d_{b,t}^{BESS}$ are storage charging and discharging, and $w_{n,t}$ is renewable wastage or curtailment. BESS inventory follows:

$$SOC_{b,t} = SOC_{b,t-1} + \eta_c c_{b,t}^{BESS} - d_{b,t}^{BESS}/\eta_d,$$

$$0 \leq SOC_{b,t} \leq z_b^E, \quad 0 \leq c_{b,t}^{BESS}, d_{b,t}^{BESS} \leq z_b^P.$$

The primary scenarios are: S0 baseline without EV load; S1 EV load with constrained transmission and no new BESS; S2 EV load with constrained transmission and endogenous BESS; and S3 EV load with relaxed transmission limits. The congestion-emissions penalty is:

$$P_{cong} = (E_{S1} - E_{S0}) - (E_{S3} - E_{S0}),$$

and BESS mitigation is:

$$M_{BESS} = (E_{S1} - E_{S0}) - (E_{S2} - E_{S0}).$$

The experiment is staged to remain computationally feasible. First, the existing balancing-area GOOD runs are used to identify seasons, hours, and regions where EV charging most strongly changes dispatch or capacity expansion. Second, the transmission-constrained model is run for a reduced set of representative weeks and candidate network aggregations. Third, the most policy-relevant cases are extended to a longer chronology or full-year run as computational time permits. The reporting metrics are designed to be comparable across these stages: binding line-hours, average congestion shadow price, renewable curtailment, fossil generation delta, BESS power and energy capacity, and emissions per kWh of EV charging. This staged design allows preliminary evidence to be submitted before the full nodal model is complete while preserving a clear path to final-paper validation.

4. Preliminary Results

The transmission-constrained model is in progress. Completed Chapter 2 runs provide two pieces of preliminary evidence. First, the charging-load generator produces structured diurnal demand, with residential evening charging and workplace/public daytime components (Fig. 1). Second, representative-week GOOD runs indicate that EV load changes consequential generation and capacity-expansion choices. Natural gas is frequently dispatched to meet induced charging demand, while optimized renewable and BESS capacity varies across representative weeks.

Before the final paper, the model will report line binding hours, congestion shadow prices, renewable curtailment, fossil generation deltas, BESS siting, and the percentage of P_{cong} closed by storage. Behavioral uncertainty will be bounded using the proxy-ranked charging-profile approach from Chapter 2: stochastic EV load profiles are scored by baseline marginal-emission exposure, and low-, median-, and high-score profiles are evaluated in the constrained model when computation allows.

5. Implications

The proposed framework distinguishes temporal emissions from delivery-constrained emissions. If BESS reduces emissions primarily by charging during renewable-rich hours and discharging during evening EV demand, storage value is temporal. If BESS reduces the gap between constrained and relaxed-transmission cases, it directly mitigates congestion-related emissions. This distinction matters for planning because managed charging, renewable expansion, BESS deployment, and transmission upgrades target different mechanisms.

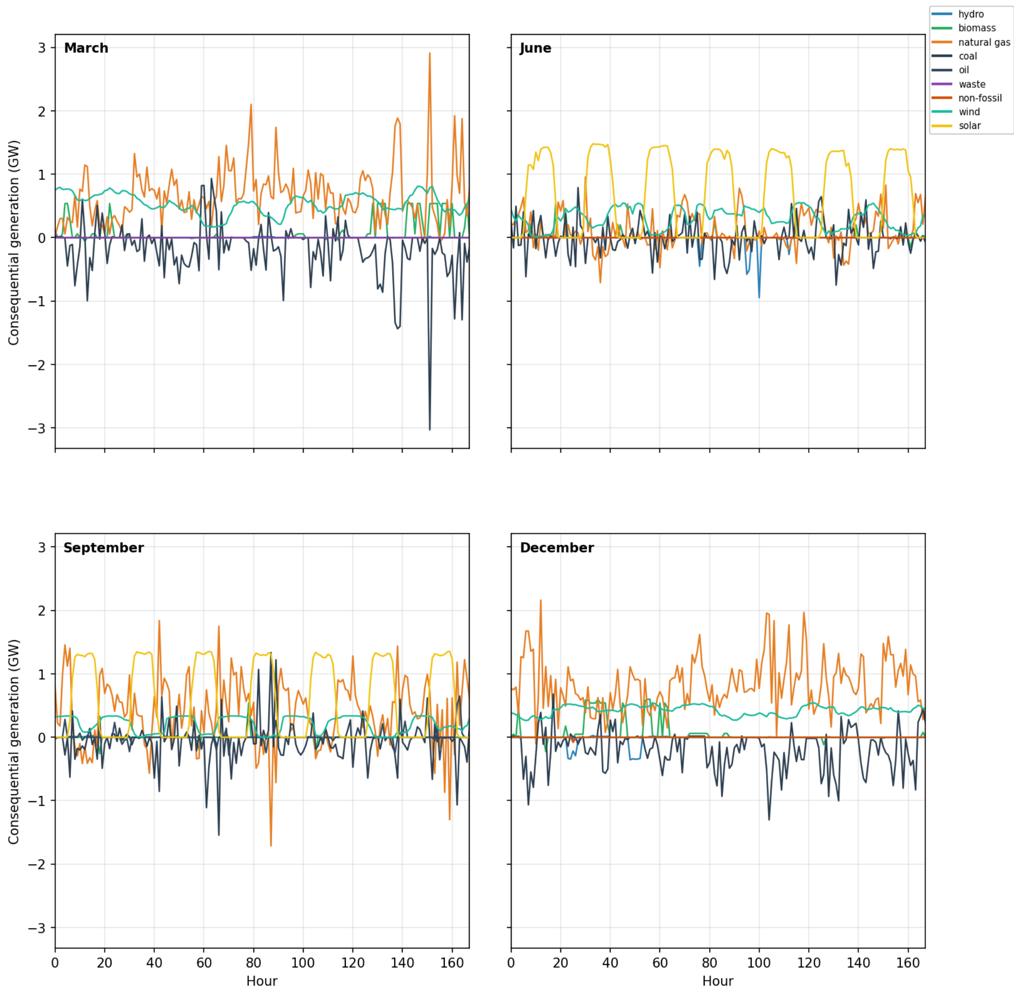

Figure 2: Preliminary representative-week consequential generation by fuel from the Chapter 2 balancing-area GOOD model. These outputs motivate the transmission-constrained extension by showing that EV load changes marginal dispatch.

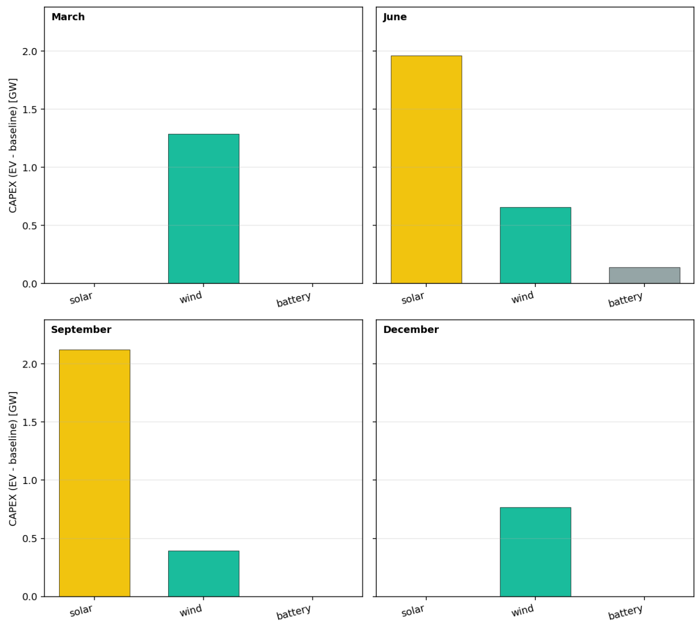

Figure 3: Preliminary representative-week capacity-expansion screening. These results will be used to set plausible CAPEX ranges for transmission-constrained BESS and renewable expansion experiments.

6. Conclusion

This extended abstract proposes a transmission-constrained extension of the dissertation's EV-grid emissions framework. It combines SOC-aware multi-day EV charging demand, empirical session timing, GOOD dispatch and capacity expansion, explicit line-flow constraints, and endogenous BESS siting. The result will be a quantified congestion-emissions penalty and a direct estimate of how much strategically deployed storage can preserve the climate benefits of EV charging under constrained power delivery.

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